

MAHAT JOSHI

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EDUCATION

University of Toronto — Engineering, Computer Engineering (CE), PEY Co-op | *Expected 2028*

Relevant Coursework: Control Systems, Linear Control Systems, Robot Modeling and Control, Probability and Applications, Operating Systems, Computer Systems Programming, Digital Signal Processing, Matrix Algebra and Optimization, Computer Security

PROJECTS

SpaceNet — Satellite Sun-Vector Astrodynamics Library

2026 – Present

C++, Python (pybind11), SOFA, CMake, GitHub Actions

- Built a satellite sun-vector library implementing two independently coded reference-frame transformation chains (classical Vallado 4-frame method and IAU-2006/2000A via SOFA), cross-validated against each other and against Vallado's published test cases to sub-0.01° accuracy
- Diagnosed a ~29-day periodic discrepancy between the two chains, tracing the root cause to an omitted lunar perturbation term consistent with quarter-moon geometry
- Designed a modular architecture in which each module trusts validated inputs and guarantees validated outputs, concentrating defensive checks at explicit interface boundaries
- Built a three-tier API (C++ core, pybind11 bindings, Python convenience layer) with a CI pipeline running 37 automated tests

Real-Time Market Simulator

ECE243 — Advanced Embedded Systems

Bare-metal C, RISC-V (Nios V), Intel DE1-SoC FPGA

- Derived and implemented a discrete-time GBM pricing engine from the continuous-time Itô SDE; hand-wrote sqrt/log/exp/sin/cos from scratch for a bare-metal target with no libm support
- Designed an interrupt-driven, no-OS architecture with timer and button ISRs synchronized via volatile shared state
- Identified and fixed 5 quantified bugs, including a Taylor-series truncation causing systematic price drift and a signed-shift undefined-behavior error

14 MHz Push-Pull RF Power Amplifier — Team Lead, RF Subsystem

ECE295

MATLAB, Analog Circuit Design

- Led the RF transmitter subsystem team in selecting an analog push-pull Class AB/B architecture over a digital approach for built-in redundancy
- Built a MATLAB simulation suite (DC bias, AC/noise, transient ODE45, parasitics) across the pre-amp, power amplifier, and cascaded chain
- Identified critical failure modes pre-hardware (thermal runaway, drain overvoltage, VHF oscillation) with quantified mitigations prior to physical bring-up

FPGA Pong — Digital Logic Implementation

ECE241

Verilog RTL, Intel DE1-SoC FPGA

- Designed an 8-state FSM governing paddle motion, gated to a frame-synchronized timing reference derived via edge detection on VGA_VS
- Implemented a two-stage synchronizer and debounce circuit for clock-domain correct pushbutton input

RESEARCH

Active Feedback Stabilization of a Continuous Z-Pinch

Independent — 2026

- Derived an MHD equilibrium model and a delayed-feedback amplitude equation for plasma boundary stability under finite sensor/actuator latency
- Decomposed the closed-loop characteristic equation into coupled stability conditions using delay-differential analysis applicable to real-time control-loop design

SKILLS

Languages: C, C++, Python, MATLAB, Verilog (RTL/structural HDL), RISC-V assembly

Domains: Astrodynamics, real-time embedded systems, digital logic and FSM design, analog RF design, control theory, numerical methods

Tools: pybind11, CMake, pytest/GoogleTest, GitHub Actions, Git, FPGA toolchains (Quartus)